HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 19 Name: _____ Marks: / 22

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let $g(x) = 13 - 5x^2$. If α is a constant, find $g(1-3\alpha)$.

   A. $8 - 45\alpha^2$

   B. $8 + 45\alpha^2$

   C. $8 - 30\alpha + 45\alpha^2$

   D. $8 + 30\alpha - 45\alpha^2$

(DSE-2023 PII Q08)

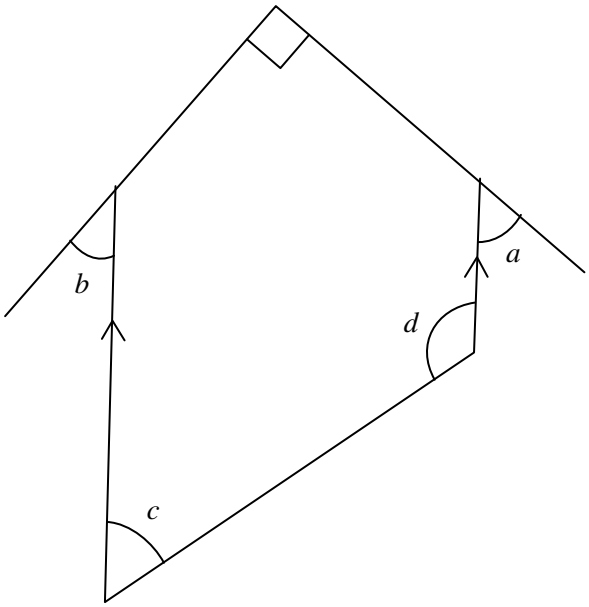
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2. According to the figure, which of the following must be true?

?



I. $a+b=90^\circ$

II. $c+d=180^\circ$

III. $a+b+c=d$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q18)

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3. It is given that $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Which of the following must be true?

I. $AE = BE$

II. $\frac{AE}{AC} = \frac{BE}{BD}$

III. $AE^2 + BE^2 = CD^2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

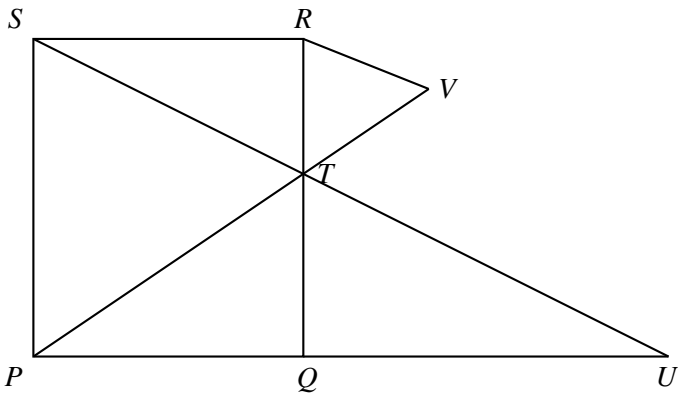
(DSE-2023 PII Q19)

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4. In the figure, $PQRS$ is a rectangle. Let T be a point lying on QR such that $\angle PTS = 90°$. PQ produced and ST produced meet at the point U. PT is produced to the point V such that $RT = RV$. Which of the following must be true?



A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

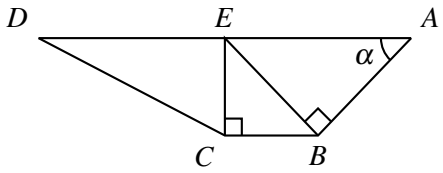
(DSE-2023 PII Q21)

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5. In the figure, $ABCD$ is a trapezium with $AD//BC$. Let E be the mid-point of AD. It is given that $\angle ABE = \angle BCE = 90°$. Find $\dfrac{CE}{DE}$.



A. $\dfrac{1}{2}$

B. 1

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

(DSE-2023 PII Q23)

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6. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

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7. The equations of the circles C_1 and C_2 are $x^2+y^2+7x-4y+15=0$ and $2x^2+2y^2-2x-16y-17=0$ respectively. Let G_1 and G_2 be the centres of C_1 and C_2 respectively. Denote the origin by O. Which of the following are true?

I. $\triangle OG_1G_2$ is an equilateral triangle.

II. The line segment OG_1 lies inside C_2.

III. C_1 and C_2 intersect at two distinct points.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q27)

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8. A box contains five cards numbered 1, 2, 3, 4 and 5 respectively while another box contains four cards numbered 6, 7, 8 and 9 respectively. If a number is randomly drawn from each box, find the probability that the product of the two numbers drawn is divisible by 4.

A. $\frac{1}{5}$

B. $\frac{3}{10}$

C. $\frac{7}{20}$

D. $\frac{9}{20}$

(DSE-2023 PII Q28)

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9. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

(DSE-2023 PII Q37)

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10. The straight line $hx + ky = 6$ and the circle $x^2 + y^2 - 8x - 4y - 18 = 0$ intersect at the points M and N, where h and k are constants. If the coordinates of the mid-point of MN are $(1, 0)$, find k.

A. 4

B. 6

C. 9

D. 12

(DSE-2023 PII Q39)

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11. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

I can finish _____ questions

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